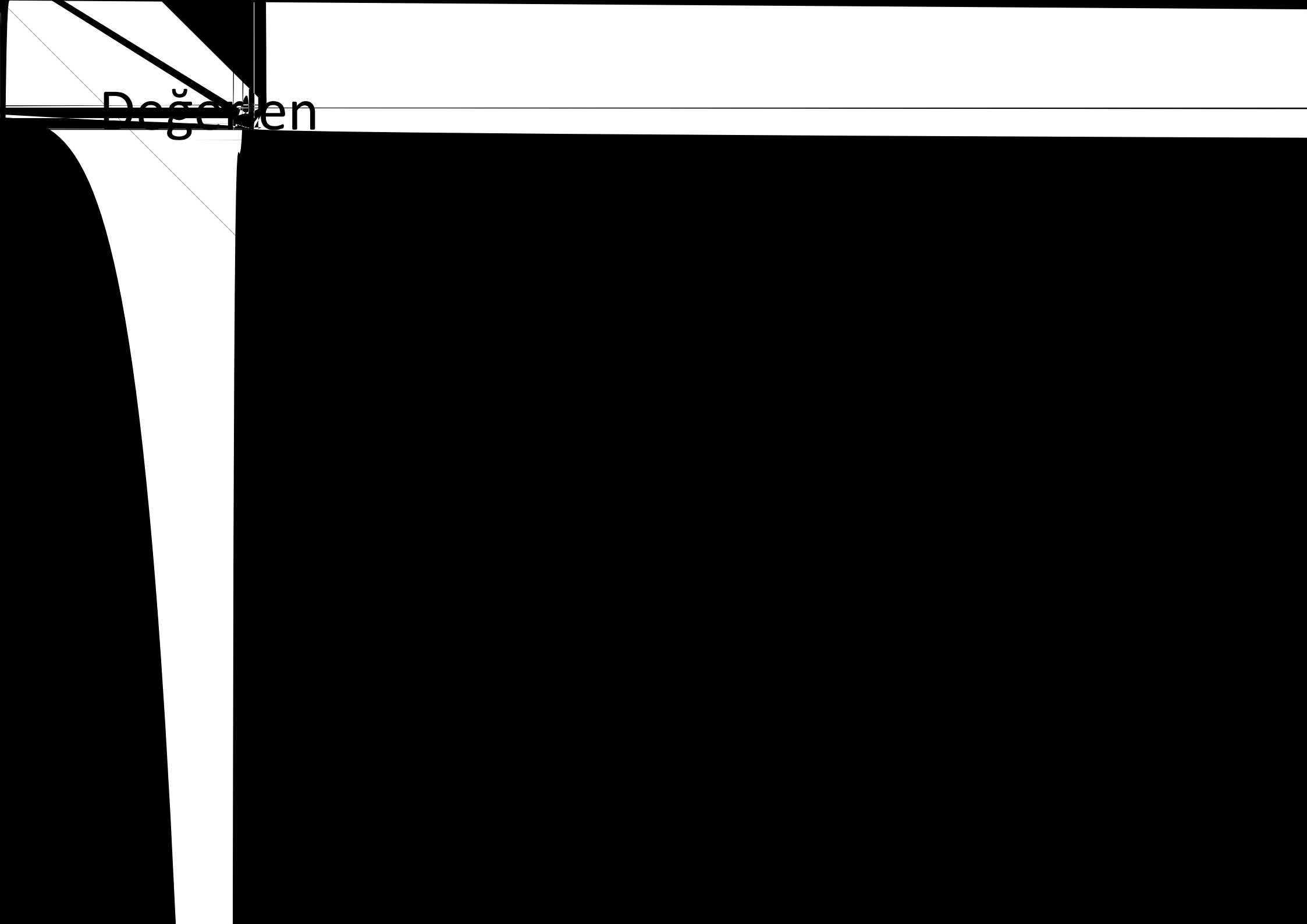
BLM 2001 BİLGİSAYAR PROGRAMLAMA 2

(PHYTON PROGRAMMING)

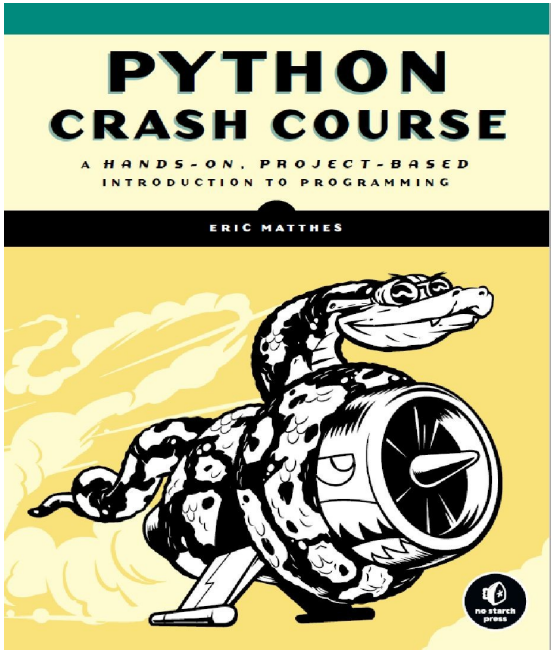
Ders Bilgisi

- Phyton programlama dersi daha önce phyton ile programlama deneyimi olan veya olmayan tüm öğrenciler içindir.
- Dersin amacı
  - problem çözme sürecinde phyton programlama dilini anlamak
  - Belirlenen hedeflere ulaşabilmek için yazılan programlarda kendine güvenilir bir şekilde bu yetiyi kazanabilmek


Değerler

Ders Kitabı ve Kaynaklar

- Python Crash Course: Eric Matthes *(No Starch Press, 2016)*
- One of the best books for learning Python



Lab Ed$_{f}$

ar

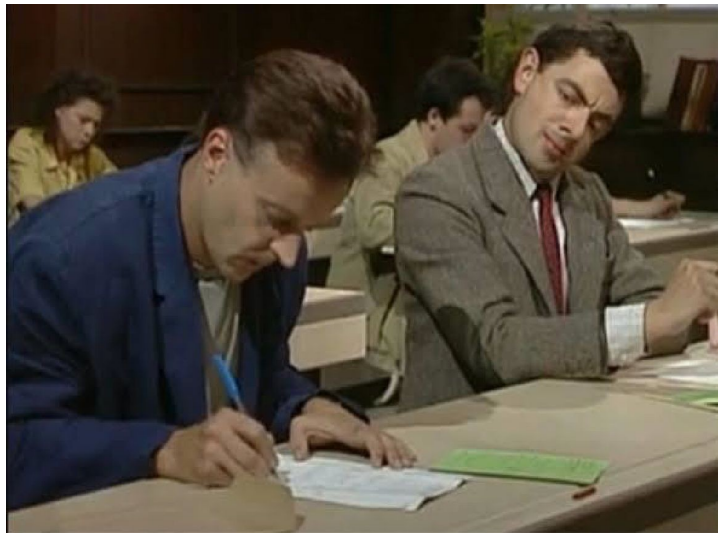
[redacted]e Kaynaklar

- Head First Python, 2nd edition Paul Barry *(O'Reilly, 2016)*
- Learn Python 3 the Hard Way Zed A. Shaw *(Addison-Wesley, 2016)*
- The official Python Tutorial
  - https://docs.python.org/3/tutorial/
- How to think like a computer scientist
  - http://openbookproject.net/thinkcs/python/english3e/
- Code Academy Python
  - http://www.codecademy.com/tracks/python

Ders Kuralları

- D

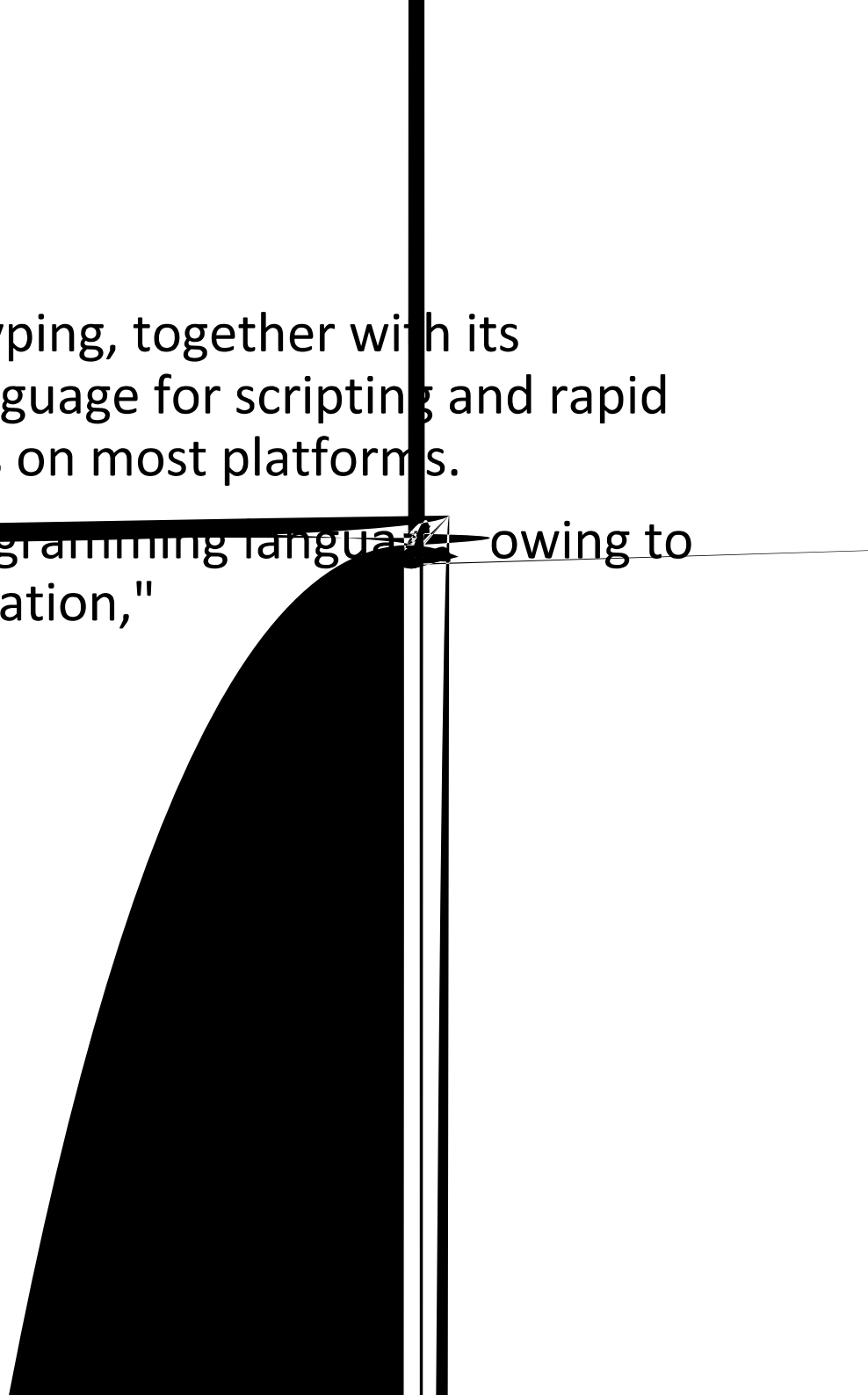
a hard deadli





Phyton Nedir?

- Python’s elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms.
- "Python is one of the most popular programming languages owing to its ease of use and wide range of application,"



Interpreter/Compiler

- Interpreter kaynak koddan 1 ifade okur, onu makine kodu veya sanal koda dönüştürür, daha sonra doğru bir şekilde execute eder.
- Compiler giriş kaynak kodu makine koduna çevirir, ardından makine kod dosyası çalıştırılır.

Compiled programs run faster than interpreted programs because they are first converted native machine code.

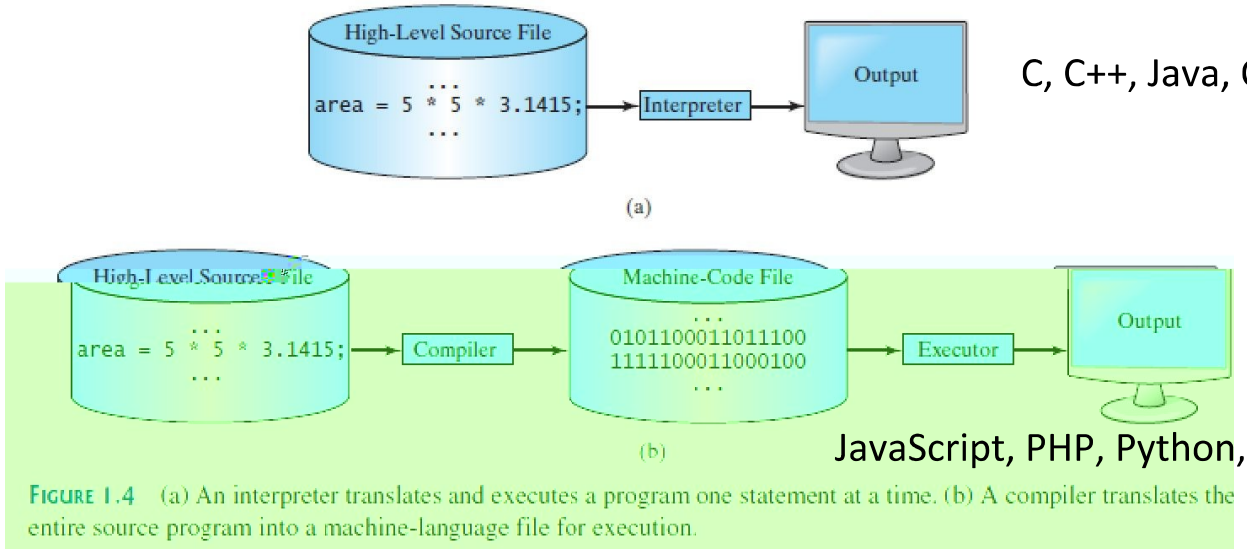


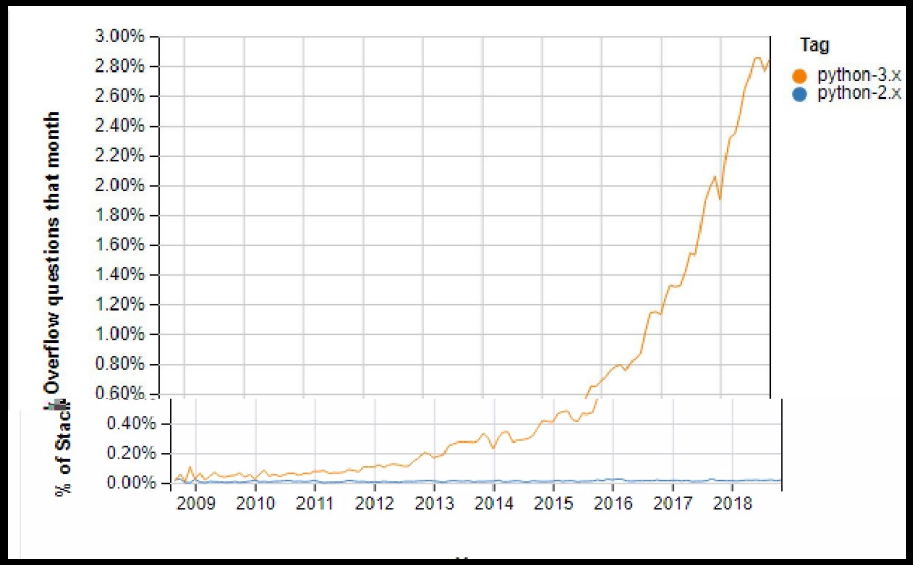
FIGURE 1.4 (a) An interpreter translates and executes a program one statement at a time. (b) A compiler translates the entire source program into a machine-language file for execution.

C, C++, Java, C#

JavaScript, PHP, Python, VBScript

Nerelerde Kullanılır?

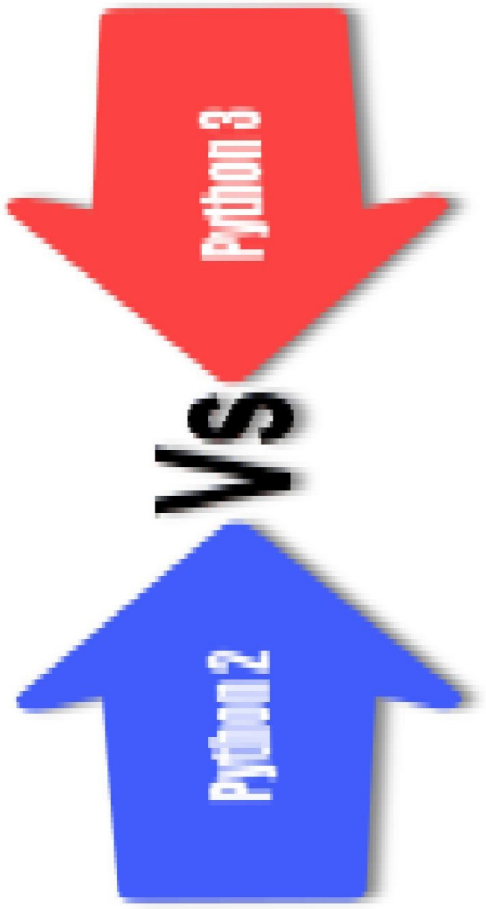
- Veritabanı Programcılığı
- GUI (Graphic User Interface) Programcılığı
- İnternet Programcılığı
- Yapay Zeka (Öğrenme Algoritmaları, Doğal Dil İşleme)


% of Stack Overflow questions that month
3.00%
2.80%
2.60%
2.40%
2.20%
2.00%
1.80%
1.60%
1.40%
1.20%
1.00%
0.80%
0.60%
0.40%
0.20%
0.00%
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
Tag
python-3.x
python-2.x

Neden Phyton3

- **INSTAGRAM**
  - In 2017, Instagram migrated the majority of their Python code base from Python 2.7 to Python 3
  - *Çünkü:*
    - Python is not traditionally a typed language, but Python v3.5 supports typing, which removes development conflicts when working new pieces of code.
    - Each newer version of Python continues to get faster runtime. Meanwhile, nobody's currently working to make Python 2.7 work faster.
    - Community support is better with Python 3.

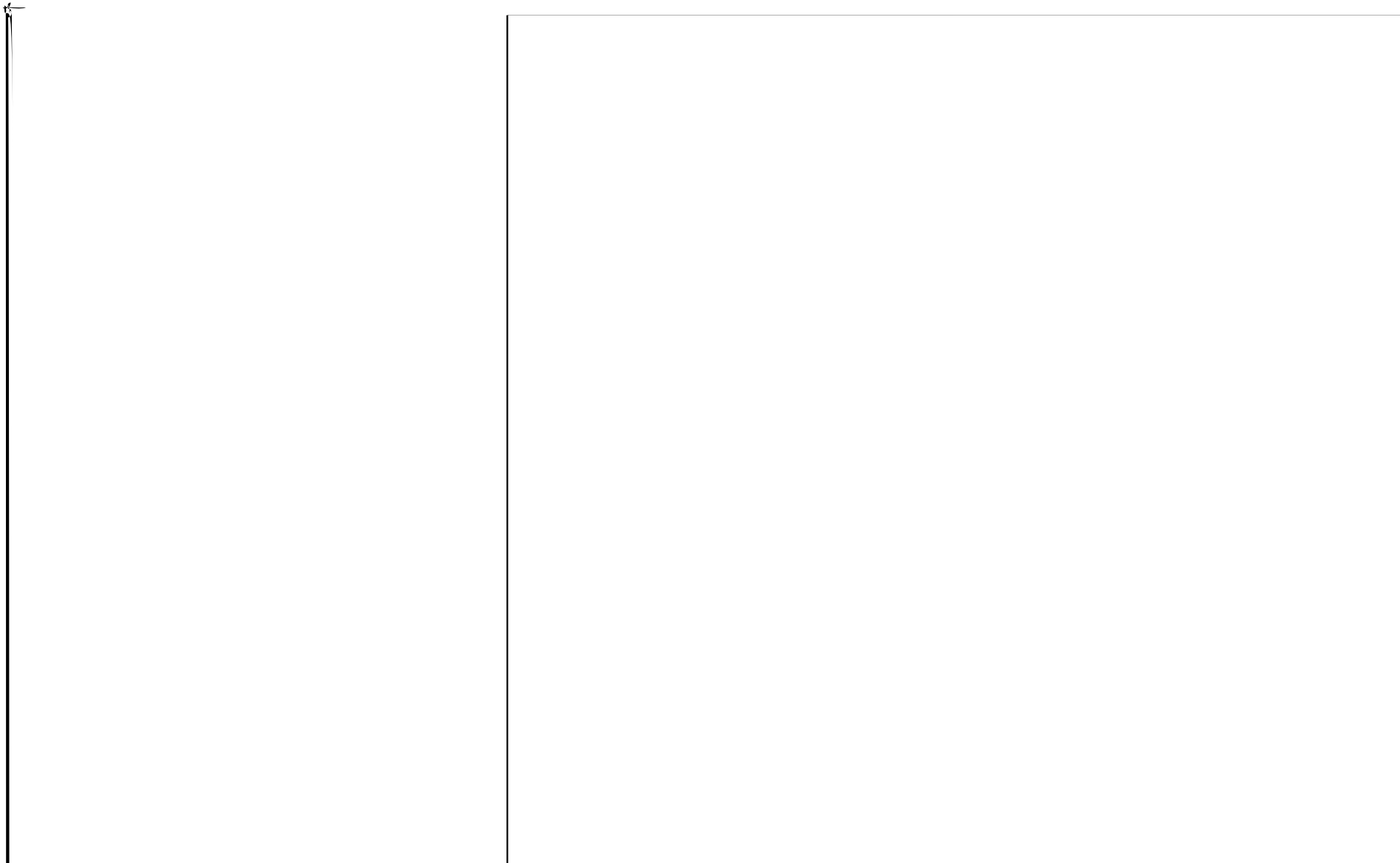
Phyton2 ile Phyton3 Farkları



Basis of comparison	Python 3	Python 2
Release Date	2008	2000
Function print	print ("hello")	print "hello"
Division of Integers	Whenever two integers are divided, you get a float value	When two integers are divided, you always provide integer value.
Unicode	In Python 3, default storing of strings is Unicode.	To store Unicode string value, you require to define them with "u".
Syntax	The syntax is simpler and easily understandable.	The syntax of Python 2 was comparatively difficult to understand.
Rules of ordering Comparisons	In this version, Rules of ordering comparisons have been simplified.	Rules of ordering comparison are very complex.
Iteration	The new Range() function introduced to perform iterations.	In Python 2, the xrange() is used for iterations.

Tarihçesi

- **Python 2**
  - Python 2.0 - October 16, 2000
  - Pyth

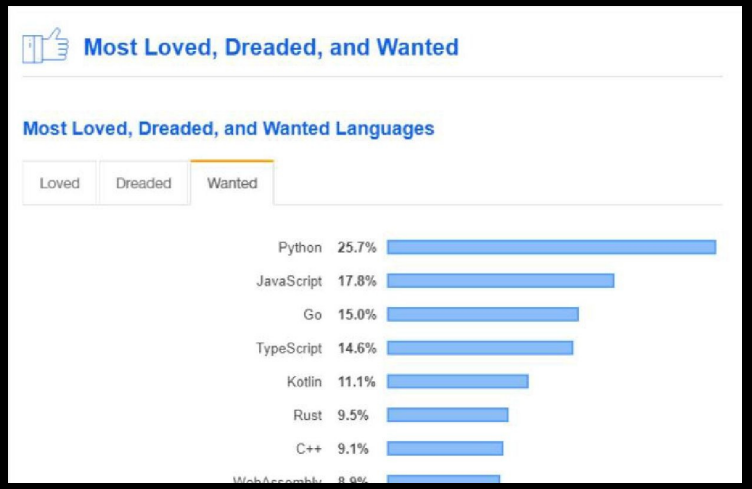


Phyton2 ne zaman öğrenilir.

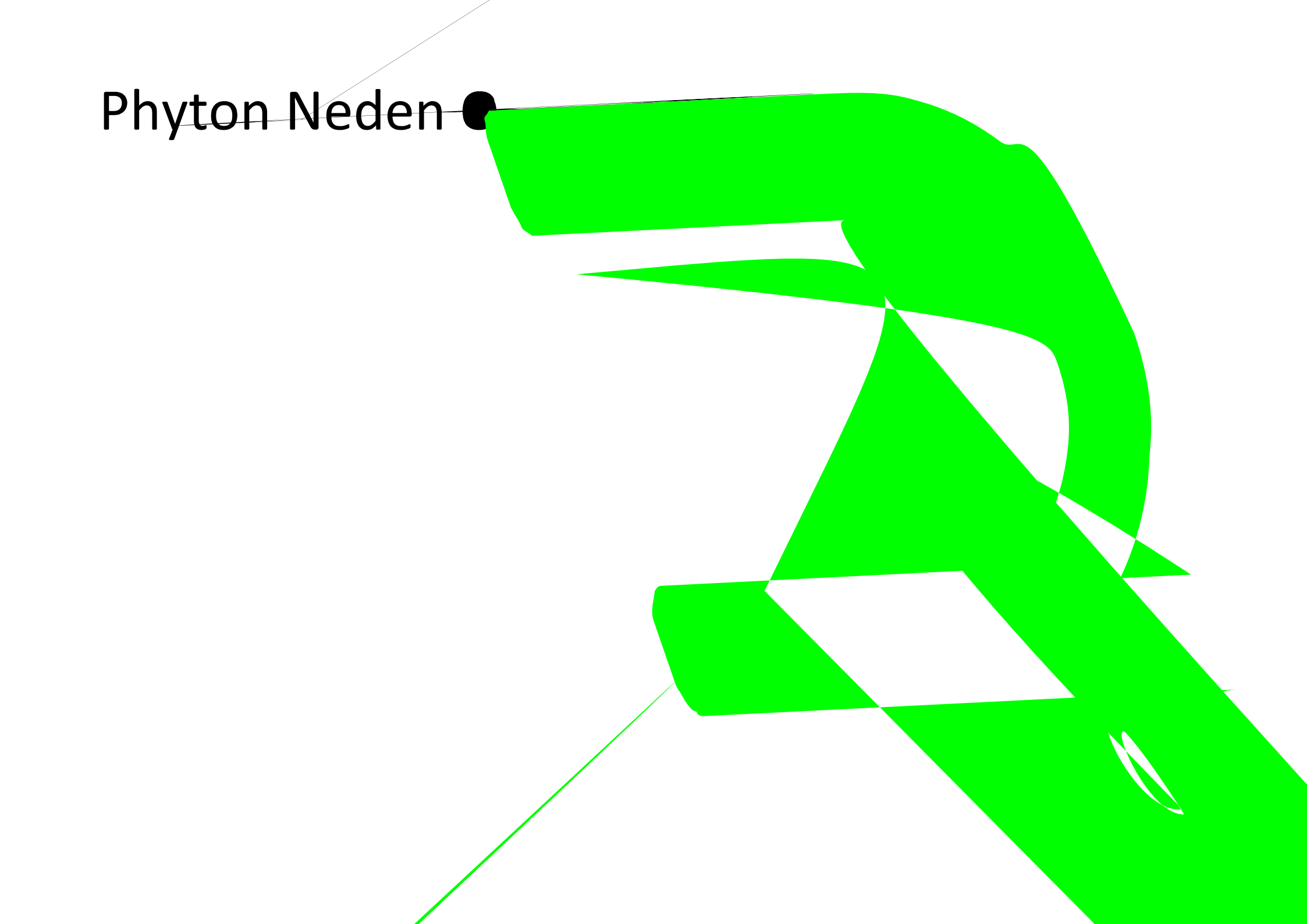
- Although, Python 2 is an old open source version here are where you still need to learn Python 2:
  - If your company's code written in Python 2, you will require to learn to work with that
  - If your development team is working on a project that depends on specific third-party libraries or software wh

e only option avail

Aug 2021	Aug 2020	Change	Programming language	Ratings	Change
1	1		C	12.57%	-4.41%
2	3	↑	Python	11.86%	+2.17%
3	2	↓	Java	10.43%	-4.00%
4	4		C++	7.36%	+0.52%
5	5		C#	5.14%	+0.46%
6	6		Visual Basic	4.67%	+0.01%
7	7		JavaScript	2.95%	+0.07%
8	9	↑	PHP	2.19%	-0.05%
9	14	↑↑	Assembly language	2.03%	+0.99%
10	10		SQL	1.47%	+0.02%
11	18	↑↑	Groovy	1.36%	+0.59%
12	17	↑↑	Classic Visual Basic	1.23%	+0.41%
13	42	↑↑	Fortran	1.14%	+0.83%
14	8	↓↓	R	1.05%	-1.75%
15	15		Ruby	1.01%	-0.03%
16	12	↓↓	Swift	0.98%	-0.44%
17	16	↓	MATLAB	0.98%	+0.11%
18	11	↓↓	Go	0.90%	-0.52%
19	36	↑↑	Prolog	0.80%	+0.41%
20	13	↓↓	Perl	0.78%	-0.33%

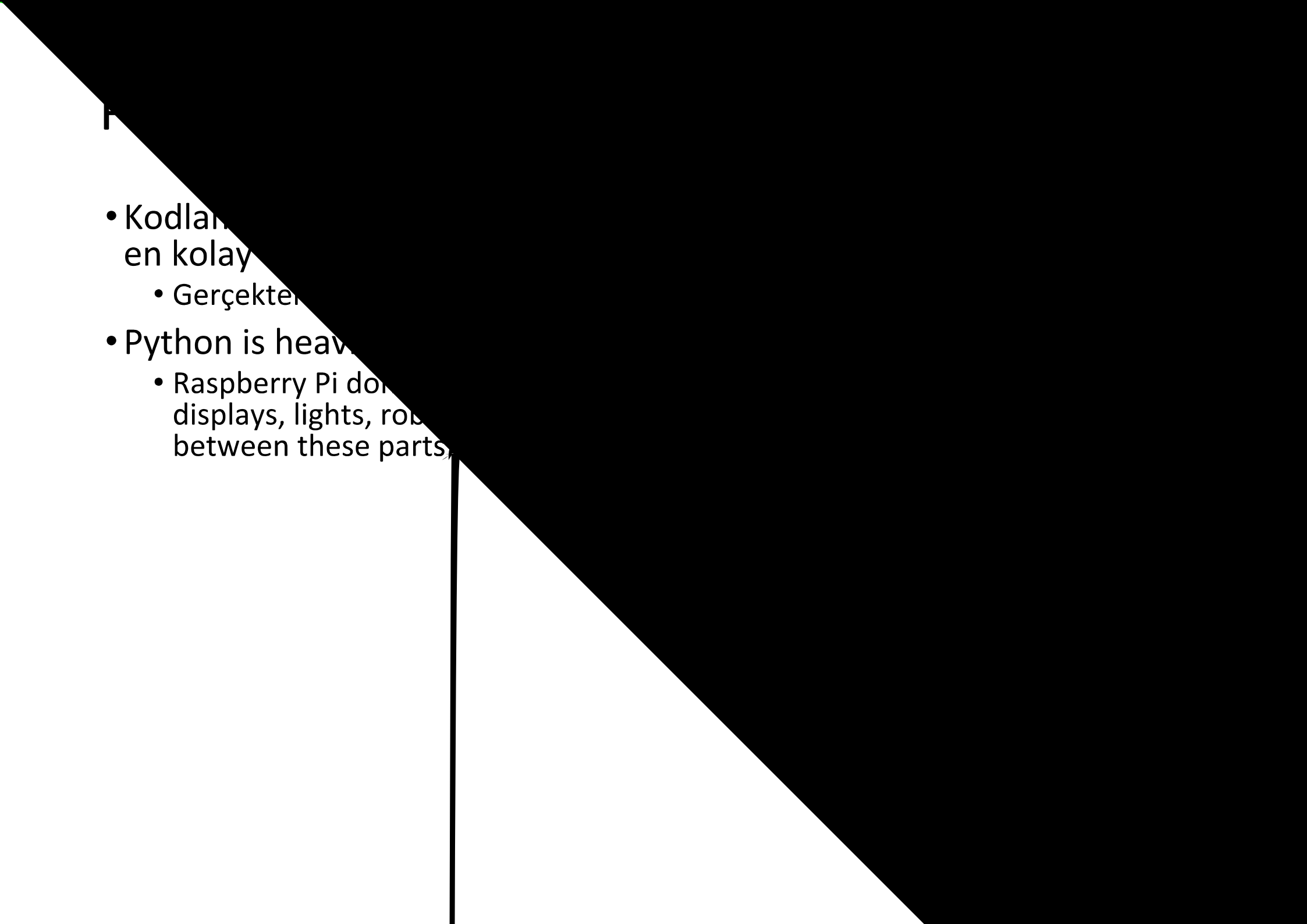


Phyton Neden



P

- Kodlam
  en kolay
    - Gerçekte
- Python is heav
    - Raspberry Pi do
      displays, lights, ro
      between these parts,



Phyton Programcılarının Maaşları

State	Python Developer Salary in 2021
Maine	**$136,778**
New York	$133,526
Delaware	$130,50
Massachusetts	$126,460
Washington	$122,022
Washington DC	$120,029
Illinois	$119,902
Connecticut	$119,845
Virginia	$119,67
South Carolina	$117,980
Texas	$117,436
Minnesota	$117,230
Florida	$116,838
Arizona	$116,658
California	$115,988
New Hampshire	$115,792
Ohio	$114,732
Tennessee	$114,640

According to Ziprecruiter, Average Python programmer salary in the world in 2021 is **$111,899** per year (or **$54 per hour**). Entry Level Python programmer salary in the world in 2021 is **$80,499** per year (or **$39 per hour**). Backend Python developers earn **$126,499 per year** (or **$61 per hour**). When Full Stack Python developers are paid on average **$128,150** per year (or **$62 per hour**).

Country	Python developer salary
The USA	**$108,598**
Switzerland	$103,55
Norway	$70,590
Denmark	$78,770
United Kingdom	$54,090
Sweden	$54,780
Germany	$65,020
Netherlands	$57,120
Finland	$56,680
Australia	$61,980
France	$50,730
Austria	$57,510

- message= "H

- Output: Hello Python world!

İsimlendirme ve Değişkenlerin Kullanımı

Phyton'da değişken kullanımında, bazı kurallar ve yönlendirmeler bulunmaktadır:

Hataya Neden Olanlar:

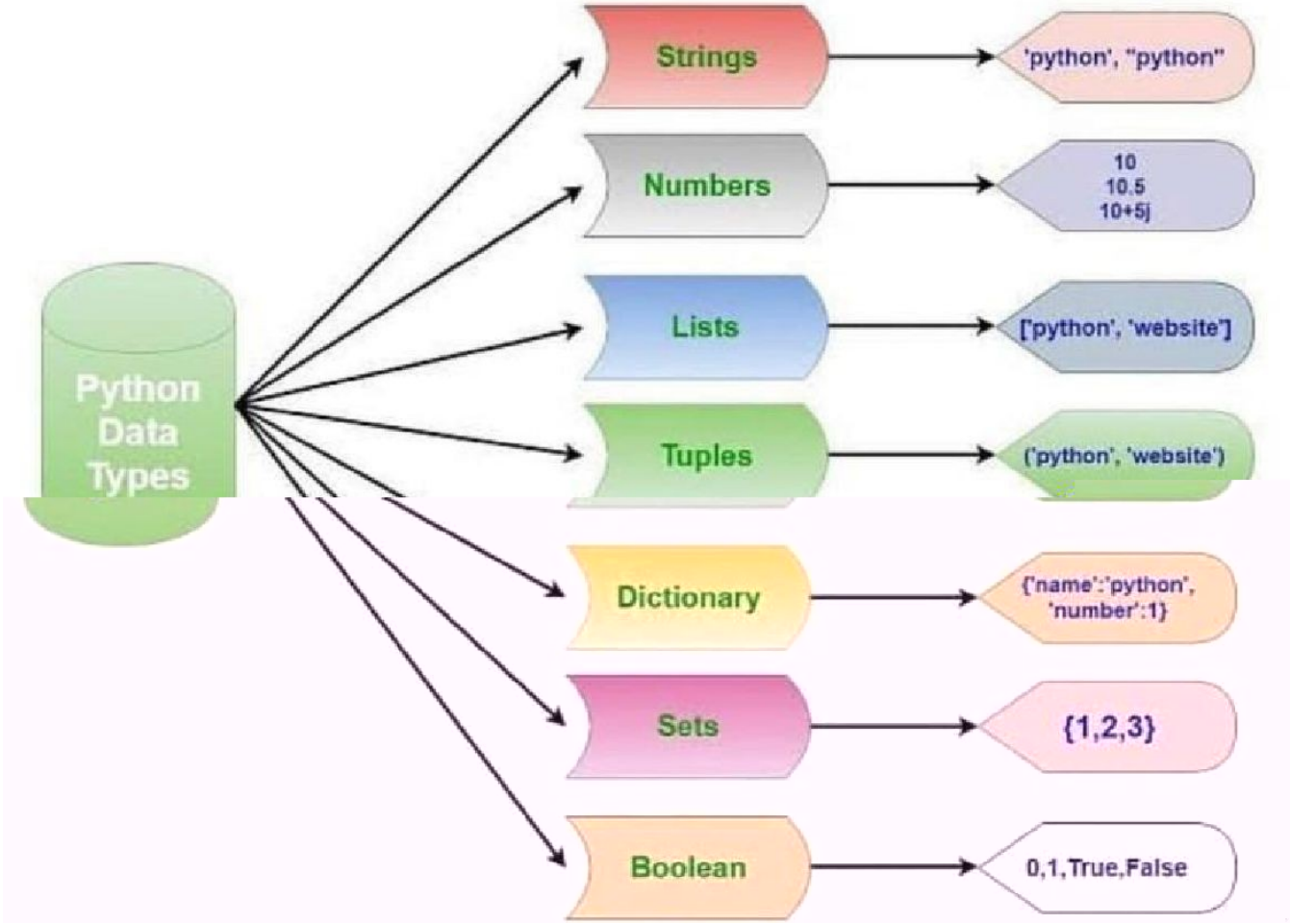
- Değişken isimlerinin sadece harf,rakam ve alt çizgi içermesi
- Harf veya altçizgi ile başlayabilir, ama sayı ile başlamaz.
- Boşluklar değişken adlarında izin verilmez, ama değişken isimlendirmede alt çizgi kullanılabilir
- Phyton daki keywordler ve limeler değişken isimlerinde kullanılamaz

i kısa ve tanımlayıcı olmalı.

-

[redacted]

Phyton Veri Tipleri



n ipleri: String

ka serisidir.

herşey string olarak Phyton ul e

lso string'

glerin durumu değiştirilebilir.

tems"

()

Değişken Tipleri: String

- Combining or Concatenating Strings
  - first_name = "lovelace“
  - full_name = first_name **+** " " **+** last_name
  - print(full_name)
  - Output: ada lovelace
- print("Hello, " **+** full_name.title() **+** "!")
  - Output : Hello, Ada Lovelace!
- Concatenation:
  - message = "Hello, " **+** full_name.title() **+** "!“
  - print(message)

Değiₚ

rint("\tPython")

- print("Languages:\nPython\nC\nJavaScript")
- print("Languages:\n\tPython\n\tC\n\tJavaScript")

Languages:

Python

C

JavaScript

Değişken Tipleri: String

- ‘phyton’ ve ‘phyton ’
- İkisi farklı
- Phyton ekstra boşluk için stringin sağını ve solunu kontrol eder.
  - Strip()/lstrip()/rstrip().

```
>>> favorite_language = 'python '
>>> favorite_language
'python '
>>> favorite_language.rstrip()
'python'
>>> favorite_language
'python '
```

value of favorite_language ?

Değişken Tipleri: String

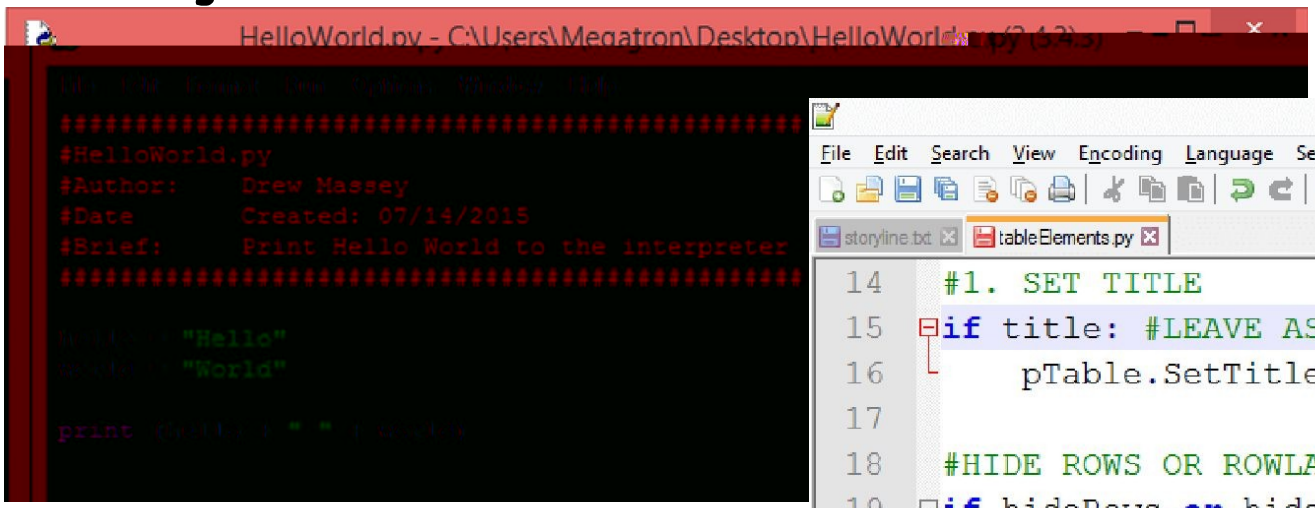
- To remove the whitespace from the string permanently ?

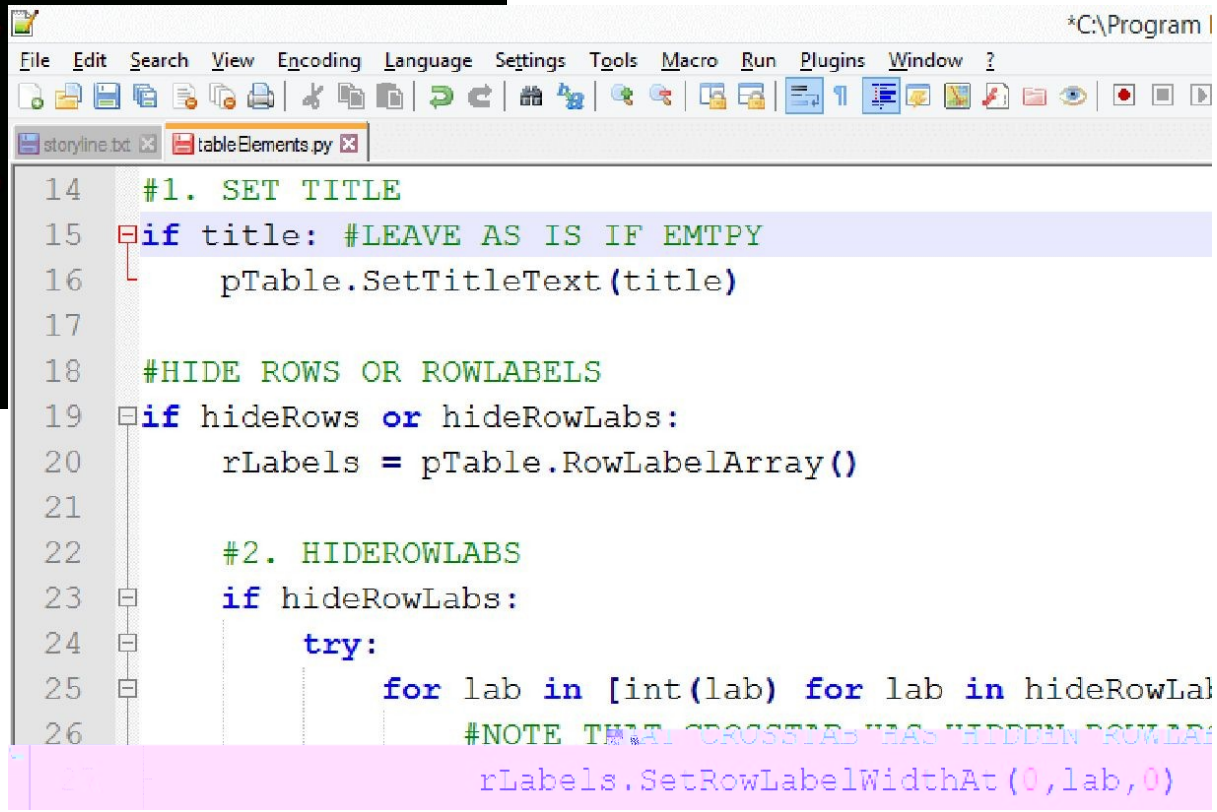
```
>>> favorite_language = 'python '
>>> favorite_language = favorite_language.rstrip()
>>> favorite_language
'python'
```

Değişken Tipleri: Str



Açıklama Satırları Örnekleri





Teşekkürler.